

Nuclear Problem Sheet

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1 Introduction

The mass of an atomic nuclei is always less than the total mass of the individual protons and neutrons. The mass of a nucleus is the binding energy subtracted from the mass of its individual constituents (protons and neutrons). The binding energy is the energy required to break an atomic nucleus into constituent protons and neutrons. The energy is related to mass due to Einstein's famous mass-energy equivalence equation, $E = mc^2$ [1].

$$M_x = N_p M_p + N_n M_n - \frac{B}{c^2} \quad (1)$$

In a nuclear reaction, the energy released in the form of thermal energy is the difference in the sums of binding energies [1]. The mass is conserved on both sides of the equation, so this does not have an effect on the energy released.

$$E = \Sigma B_p - \Sigma B_r \quad (2)$$

(B_p is the total binding energy of the products and B_r is the total binding energy of the reactants)

We can use the Semi-Empirical Mass Formula to predict the binding energy of an atomic nucleus based on the protons and neutrons, by treating the nucleus like a "liquid-drop". The formula is given below [1].

$$B(A, Z) = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_a \frac{(A - 2Z)^2}{A} + \delta(A, Z) \quad (3)$$

$$a_v \approx 15.56 \text{ MeV} \quad (\text{Volume coefficient})$$

$$a_s \approx 17.23 \text{ MeV} \quad (\text{Surface coefficient})$$

$$a_c \approx 0.711 \text{ MeV} \quad (\text{Coulomb coefficient})$$

$$a_a \approx 23.702 \text{ MeV} \quad (\text{Asymmetry coefficient})$$

$$a_p \approx 34.0 \text{ MeV} \quad (\text{Pairing coefficient})$$

The Volume term represents the strong nuclear force that holds the nucleons together.

The Surface term corrects for the nucleons on the outer edge of the nucleus.

The Coulomb term represents electrostatic repulsion between protons.

The Asymmetry term accounts for the Pauli-Exclusion Principle.

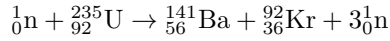
The Pairing term accounts for the spin-coupling of neutron pairs. An even number of both protons and neutrons gets a positive energy boost. An odd number of both protons and neutrons gets a negative effect, and an odd-even pairing has no net effect [2].

2 Fission vs Fusion

2.1 Fission - Uranium

The primary method for nuclear fission uses Uranium as fuel [3].

The nuclear reaction of the fission of Uranium-235 is [1]:



Using equation (1) we can calculate the mass of Krypton-92, which has 36 protons and 56 neutrons.

$$M_x = 36 \times 1.67 \times 10^{-27} + 56 \times 1.67 \times 10^{-27} - \frac{764.77 \times 10^6 \times 1.60 \times 10^{-19}}{(3.00 \times 10^8)^2}$$

$$M_x = 1.52 \times 10^{-25} \text{kg}$$

$$M_x = 92\text{u}$$

Using equation (2) we can calculate the binding energy of Uranium.

$$E = B_{Ba} + B_{Kr} - B_U$$

The neutrons have no binding energies as they are separate particles.

$$B_U = B_{Ba} + B_{Kr} - E$$

The binding energy of Ba-141 is 1145.36 MeV, the binding energy of Kr-92 is 764.77 MeV and the nuclear reaction releases 173.28 MeV [1].

$$B_U = 1145.36 + 764.77 - 173.28$$

$$B_U = 1736.85 \text{ MeV}$$

To calculate the energy released per kilogram, we divide the Energy released in one nuclear reaction by the mass of Uranium.

$$\text{Energy/Kg} = \frac{173.28 \times 10^6 \times 1.60 \times 10^{-19}}{235 \times 1.66 \times 10^{-27}}$$

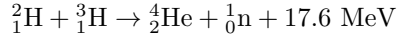
$$\text{Energy/Kg} = 7.11 \times 10^{13} \text{J kg}^{-1}$$

Natural gas has an energy density of $4.9 \times 10^7 \text{J kg}^{-1}$, which makes Uranium more than 1 million times more energy dense. The UK's total annual energy consumption is roughly 4.9×10^7 [1]. Around 140 tonnes of Uranium per year would be required to meet the UK's energy consumption needs.

2.2 Fusion

The primary method of replicating nuclear fusion on Earth (and also the most frequent nuclear reaction in the sun) is fusing deuterium and tritium into Helium [4].

The nuclear reaction for the fusion of deuterium and tritium is:



Using equation (1) we can calculate the binding energy of deuterium, which has 1 proton, 1 neutrons and a mass of 2.01u [1].

$$B_D = (N_p M_p + N_n M_n - M_x) c^2$$

$$B_D = (1 \times 1.67 \times 10^{-27} + 1 \times 1.67 \times 10^{-27} - 2.01 \times 1.66 \times 10^{-27}) \times (3 \times 10^8)^2$$

$$B_D = 1.91 \text{ MeV}$$

Using equation (2) we can calculate the binding energy of tritium (H-3).

$$E = B_{He} - B_D - B_T$$

$$B_T = B_{He} - B_D - E$$

The binding energy of He-4 is 27.27 MeV, the binding energy of deuterium is 1.91 MeV and the nuclear reaction releases 17.59 MeV [1].

$$B_T = 27.27 - 1.91 - 17.59$$

$$B_T = 8.05 \text{ MeV}$$

To calculate the energy released per kilogram of Hydrogen fuel containing deuterium and tritium, we divide the Energy released in one nuclear reaction by the mass of deuterium and tritium.

$$\text{Energy/Kg} = \frac{17.59 \times 10^6 \times 1.60 \times 10^{-19}}{(2.01 + 3.02) \times 1.66 \times 10^{-27}}$$

$$\text{Energy/Kg} = 3.37 \times 10^{14} \text{J kg}^{-1}$$

Around 30 tonnes of Hydrogen per year would be required to meet the UK's energy consumption needs.

For two nuclei to fuse they need to get within nuclear distances (around 10^{-15}m) [1]. Using the assumption that they fuse when the electrostatic potential energy of the two protons is equal to their kinetic energy due to random motion, we can calculate the temperature required to initiate fusion in a reactor on Earth.

The electrostatic potential energy can be calculated using the Coulomb potential energy equation [5]:

$$U = \frac{q_1 q_2}{4\pi\epsilon_0 r} \quad (4)$$

Where ϵ_0 is the permittivity of free space.

Therefore, the potential energy of two protons is:

$$U = \frac{e^2}{4\pi\epsilon_0 r}$$

The average kinetic energy of a single molecule is [6]:

$$U = \frac{3}{2}k_B T \quad (5)$$

To get the kinetic energy of the molecules to equal the potential energy, we will need a temperature of:

$$T = \frac{e^2}{6\pi\epsilon_0 k_B r}$$

$$T = \frac{(1.6 \times 10^{-19})^2}{6\pi \times 8.85 \times 10^{-12} \times 1.38 \times 10^{-23} \times 10^{-15}}$$

$$T = 1.1 \times 10^{10} \text{ K}$$

This is roughly 11 billion Kelvin, much larger than the temperature of the Sun, 15 million Kelvin. There is a difference in temperature because it is easier to bring the protons into close proximity due to the high pressure [4]. Additionally, although the energy can be calculated using the temperature of 15 million Kelvin, this is only the average energy. There will be protons with much higher energies [6]. Therefore, fusion is still the main energy source of solar radiation.

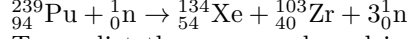
2.3 Fission - Plutonium

Using equation (3) we can predict the binding energy of Plutonium-239, which contains 94 protons and 145 neutrons. As there is an even number of protons and an odd number of neutrons, the Pairing term is equal to 0.

$$B(239, 94) = 15.56 \times 239 - 17.81 \times 239^{2/3} - 0.711 \times \frac{94^2}{239^{1/3}} - 23.702 \times \frac{(239 - 2 \times 94)^2}{239} + 0$$

$$B(239, 94) = 1763 \text{ MeV}$$

An example of a fission reaction of Plutonium is:



To predict the energy released in this reaction, we first use the SEMF to predict the binding energy of Zirconium-103 (40 protons and 63 neutrons). As there is an even number of protons and odd number of neutrons, the pairing term becomes 0.

$$B(103, 40) = 15.56 \times 103 - 17.81 \times 103^{2/3} - 0.711 \times \frac{40^2}{103^{1/3}} - 23.702 \times \frac{(103 - 2 \times 40)^2}{103}$$

$$B(103, 40) = 867 \text{ MeV}$$

$$E = B_{Xe} + B_{Zr} - B_{Pu}$$

The binding energy per nucleon for Xe-134 is 8.4137 MeV, the binding energy of Zr-103 is 867 MeV and the binding energy for Pu-239 is 1763 MeV [1].

$$E = 8.41 \times 134 + 867 - 1763$$

$$E = 231 \text{ MeV}$$

The energy released per Kilogram is:

$$\text{Energy/Kg} = \frac{231 \times 10^6 \times 1.60 \times 10^{-19}}{239 \times 1.66 \times 10^{-27}}$$

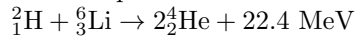
$$\text{Energy/Kg} = 9.32 \times 10^{13} \text{ J kg}^{-1}$$

The energy released per kilogram of fuel of plutonium is slightly higher than the energy released per kilogram of Uranium fuel.

2.4 Aneutronic Fusion

Aneutronic fusion is fusion without the release of neutron radiation, which has the potential to be hazardous. This is because neutrons are highly penetrating due to their neutral charge, so can penetrate deeply into matter and even damage living tissue [7].

An example of aneutronic fusion is the fusion of deuterium and lithium-6.



We can use the SEMF to predict the binding energy of lithium-6. As there is an odd number of both protons and neutrons, the pairing term becomes $-a_p A^{-3/4}$.

$$B(6, 3) = 15.56 \times 6 - 17.81 \times 6^{2/3} - 0.711 \times \frac{3^2}{6^{1/3}} - 23.702 \times \frac{(6 - 2 \times 3)^2}{6} - 34.0 \times 6^{-3/4}$$

$$B(6, 3) = 22.2 \text{ MeV}$$

Using this we can predict the energy released in this fusion reaction.

$$E = 2B_{He} + B_D - B_{Li}$$

The binding energy per nucleon for He-4 is 27.27 MeV, the binding energy of Deuterium is 1.91 MeV and the binding energy for Li-6 is 22.2 MeV [1].

$$E = 2 \times 27.27 - 1.91 - 22.2$$

$$E = 30.4 \text{ MeV}$$

This is greater than the actual energy released, which is 22.4 MeV, a 36% uncertainty. This is because our value for the binding energy of Lithium was not accurate as both inputs A and Z were small [1].

This reaction is even more challenging to replicate in a fusion reactor because lithium has three protons, and will therefore experience roughly 3 times more repulsion due to Coulomb's law. This would mean that the kinetic energy of the molecules will have to be 3 times greater, so the temperature in the fusion reactor will be 3 times higher.

2.5 Which element has the highest binding energy per nucleon?

Using the SEMF we can estimate which Z value maximizes $\frac{B}{a}$.

$$B(A, Z)/A = a_v - \frac{a_s}{A^{1/3}} - a_c \frac{Z^2}{A^{4/3}} - a_a \frac{(A - 2Z)^2}{A^2} + \frac{\delta(A, Z)}{A} \quad (6)$$

We ignore the pairing term and assume that the nucleus has the same number of protons and neutrons ($A = 2Z$), although generally the number of neutrons exceed the number of protons.

$$\frac{B}{A} = a_v - a_s 2^{-1/3} Z^{-1/3} - a_c 2^{-4/3} Z^{2/3}$$

We can differentiate this and set it equal to 0 to find the Z value which maximizes $\frac{B}{a}$.

$$\frac{d(B/A)}{dZ} = \frac{1}{3} a_s 2^{-1/3} Z^{-4/3} - \frac{2}{3} a_c 2^{-4/3} Z^{-1/3}$$

$$\frac{1}{3}a_s 2^{-1/3} Z^{-4/3} = \frac{2}{3}a_c 2^{-4/3} Z^{-1/3}$$

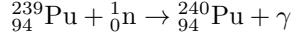
$$\frac{a_s}{2a_c} \times 2 = Z$$

$$Z = \frac{a_s}{a_c} = \frac{17.81}{0.711} \approx 25$$

The element with atomic number 25 is Manganese.

3 Wavelength of the Gamma Ray produced

In the fission of Plutonium, a side reaction may occur.



This reaction produces gamma ray radiation instead of neutron radiation, and the fission of Plutonium does not occur. This reaction occurs 27% of the number of times the fission reaction of Plutonium occurs. We can predict the minimum wavelength of the gamma rays produced using the SEMF and binding energies.

We have calculated the binding energy of Plutonium-239 as 1763 MeV. The approximate binding energy of Plutonium-240 can be calculated using the SEMF, where the pairing term is $-a_p A^{-3/4}$, because both the number of protons and neutrons are even.

$$B(240, 94) = 15.56 \times 240 - 17.81 \times 240^{2/3} - 0.711 \times \frac{94^2}{240^{1/3}} - 23.702 \times \frac{(240 - 2 \times 94)^2}{240} + 34 \times 240^{-3/4}$$

$$B(240, 94) = 1769 \text{ MeV}$$

If all the excess binding energy is transferred to a single photon, then we will obtain the smallest wavelength.

$$E = B_{240} - B_{239} = 1769 - 1763 = 6.0 \text{ MeV}$$

This is the energy of the gamma ray, which we can use to calculate its wavelength.

$$E = \frac{hc}{\lambda} \tag{7}$$

$$\lambda = \frac{hc}{E}$$

$$\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{6.0 \times 1.6 \times 10^{-13}}$$

$$\lambda = 2.1 \times 10^{-13} \text{ m}$$

4 Binding Energy per Nucleon

We can use the Semi-Empirical Mass Formula to plot the binding energy per nucleon against atomic number. The code used for the plot is given in the appendix.

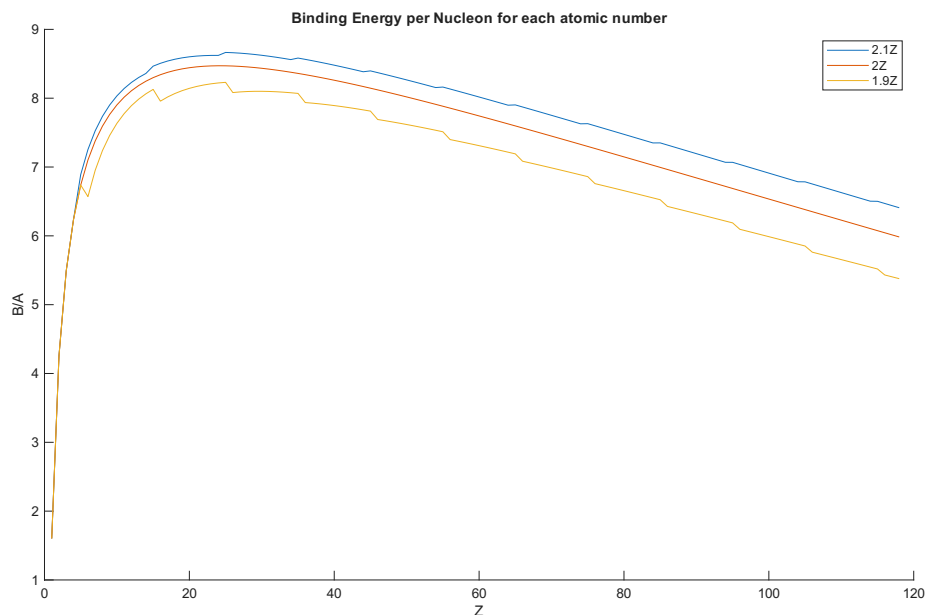


Figure 1: Plot of B/A against Z

This binding energy per nucleon increases towards Fe-56, which shows us how iron is the most tightly bound nucleus.

For light elements $A < 56$, the curve increases rapidly, showing how the fusion of two light nuclei results in dramatic movement up the slope and the release of energy.

For heavy elements $A > 56$, the curve gradually decreases, showing how fission of a large atomic nucleus will form more stable nuclei, moving up the slope and releasing energy [8].

5 How much fusion fuel for an Ion Thruster?

An ion-thruster is a type of electric spacecraft propulsion that turns natural gas into plasma by ionizing it and then accelerates those charged ions outward to propel the spacecraft.[9].

How much fuel would we need for a fusion-powered rocket that provides an equivalent thrust of the 43.7kN used in the Apollo spacecraft [1]?

Using a deuterium-tritium fusion reaction, we assume that all 17.59 MeV of energy is transferred to the kinetic energy of the helium-4 nuclei. We can use the kinetic energy to calculate the velocity.

$$E_k = \frac{1}{2}mv^2 \quad (8)$$

$$v = \sqrt{\frac{2E}{m}}$$

$$v = \sqrt{\frac{2 \times 17.59 \times 1.6 \times 10^{-19} \times 10^6}{4 \times 1.66 \times 10^{-27}}}$$

$$v = 2.9 \times 10^7 \text{ m s}^{-1}$$

We can use the velocity to calculate the momentum.

$$p = mv \quad (9)$$

$$p = (4 \times 1.66 \times 10^{-27}) \times 2.9 \times 10^7$$

$$p = 1.9 \times 10^{-19} \text{ kg m s}^{-1}$$

We can find the thrust because force is the time derivative of momentum.

$$F = \frac{dp}{dt} \quad (10)$$

Let N be the number of helium nuclei expelled per second.

$$F = Np$$

$$N = \frac{F}{p}$$

$$N = \frac{43.7 \times 10^3}{1.9 \times 10^{-19}}$$

$$N = 2.3 \times 10^{23}$$

We can convert this to fuel consumption.

$$\text{Fuel Consumption} = Nm$$

$$\text{Fuel Consumption} = (2.3 \times 10^{23}) \times (5 \times 1.66 \times 10^{-27})$$

$$\text{Fuel Consumption} = 1.9 \times 10^{-3} \text{kg s}^{-1}$$

In contrast, the Apollo spacecraft had a fuel consumption rate of 10.84 kg s^{-1} (excluding oxidizer) [10].

References

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- [4] S. A. Holgate, *Nuclear Fusion: The Race to Build a Mini-Sun on Earth*. London: Icon Books, 2022.
- [5] ScienceDirect, “Coulomb potential - an overview.” <https://sciencedirect.com>, 2026. Accessed: 09 August 2026.
- [6] Save My Exams, “6.6.3 average molecular kinetic energy.” <https://savemyexams.com>, 2026. Accessed: 09 August 2026.
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- [8] Save My Exams, “11.3 binding energy per nucleon graph.” <https://savemyexams.com>, 2026. Accessed: 09 August 2026.
- [9] The Lee Company, “Hall effect vs. ion thruster electric propulsion explained.” <https://theleeco.com>, 2026. Accessed: 09 August 2026.
- [10] RocketISP, “Rocketisp examples and documentation.” <https://readthedocs.io>, 2026. Accessed: 09 August 2026.

A Binding Energy per Nucleon MATLAB Plot Script

```
Z = linspace(1,118,118);
A_1 = round(Z.*2.1);
A_2 = round(Z.*2);
A_3 = round(Z.*1.9);
E_1 = SEMF(A_1,Z);
E_2 = SEMF(A_2,Z);
E_3 = SEMF(A_3,Z);
figure
hold on
plot(Z,E_1)
plot(Z,E_2)
plot(Z,E_3)
hold off
legend('2.1Z','2Z','1.9Z')
xlabel('Z')
ylabel('B/A')
title('Binding Energy per Nucleon for each atomic number')
hold off

function energy = SEMF(A,Z)
    volume = 15.56;
    surface = -17.23.*A.^(-1/3);
    coulomb = -0.711.*Z.^2./A.^(4/3);
    assymetry = -23.702.*(A-2.*Z).^2./A.^2;
    if mod(A,2)==0 & mod(Z,2)==0
        pairing = 34.*A.^(-7/4);
    elseif mod(A,2)==0 & mod(Z,2)==1
        pairing = -34.*A.^(-7/4);
    else
        pairing = 0;
    end
    energy = volume + surface + coulomb + assymetry + pairing;
end
```